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## RESEARCH INTEREST

My research spans two areas: Trustworthy AI and NeuroAI. Within **Trustworthy AI**, I focus on *AI Interpretability* [1, 2], probing memory traces in LLMs, *AI Steerability* [3, 4], editing and unlearning model knowledge, and *AI Alignment* [5, 6], studying human-LLM decision alignment. Within **NeuroAI**, I pursue *Neuroscience for AI* [7, 8], translating brain mechanisms into model architectures, and *AI for Neuroscience* [9, 10], applying ML to behavioral and neural data.

- [1] *AI Engram: In Search of Memory Traces in Artificial Intelligence*, **ICML 2026 Oral**, Kwon et al. (First)
- [2] *Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing*, **ICML 2026 Mech. Interp. Workshop**, Kwon et al. (First)
- [3] *Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing*, **ICLR 2026**, Kim, Kwon et al. (Co-author)
- [4] *Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning*, **ICLR 2026**, Yang, Kwon et al. (Co-author)
- [5] *Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment*, **AAAI 2026**, Kwon et al. (First)
- [6] *Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions*, **ACL 2026 Findings**, Kim, Kwon et al. (Co-corresponding)
- [7] *Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity*, **NeurIPS 2023**, Kim, Kwon et al. (First)
- [8] *Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex*, **ICLR 2025**, Kwon et al. (First)
- [9] *SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior*, **IJCV 2024**, Kwon et al. (First)
- [10] *Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination*, **Exp. Mol. Med. 2026**, Kwon et al. (First)

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## PROFESSIONAL EXPERIENCE

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|-------------------------------------------------------------------------|----------------|
| <b>Postdoctoral Researcher in Trustworthy AI</b>                        | 2024 - Present |
| Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany |                |
| ◊ Advisor: Meeyoung Cha, Scientific Director of MPI-SP                  |                |
| <b>Postdoctoral Researcher in NeuroAI</b>                               | 2022 - 2024    |
| Institute for Basic Science (IBS), Daejeon, South Korea                 |                |
| <b>Researcher in Neuroscience</b>                                       | 2018 - 2020    |
| Institute for Basic Science (IBS), Daejeon, South Korea                 |                |
| <b>Student Researcher</b>                                               | 2014 - 2018    |
| Korea Institute of Science and Technology (KIST), Seoul, South Korea    |                |

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## EDUCATION

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| <b>Ph.D. in Nano-Bio-Information-Technology</b>                            | 2014 - 2022 |
| Korea University, Seoul, South Korea                                       |             |
| ◊ Advisor: C. Justin Lee, Director of IBS                                  |             |
| ◊ Full-Ride Scholarship, KU-KIST Graduate School                           |             |
| <b>B.E. in Electrical and Electronic System</b>                            | 2008 - 2012 |
| Saitama University, Saitama, Japan                                         |             |
| ◊ Full-Ride Scholarship, Korea-Japan Governments Joint Scholarship Program |             |

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## PUBLICATION SUMMARY

### In Conferences

- *main author* (10): ICML2026, AAAI2026, ICLR2025, NeurIPS2023, ICML2026-w, ACL2026-f, ICLR2025-w, ICLR2024-w, ICLR2024-w, CVPR2023-w
- *co-author* (8): ICML2026, ACL2026, ICLR2026, ICLR2026, ICLR2026-w, ICLR2025-b, NeurIPS2022-w, CVPR2022-w

### In Journals

- *main author* (6): Exp.Mol.Med.2026, IJCV2024, Exp.Neurobiol.2024, Mol.Brain2021, Exp.Neurobiol.2018, Exp.Neurobiol.2017
- *co-author* (14): Nat.Comm.2025, Nat.Comm.2024, Nat.Metab.2023, Brain2023, Mol.Brain2022, CellMetab.2022, Exp.Neurobiol.2022, Exp.Neurobiol.2021, Exp.Neurobiol.2021, J.Neurosci.2020, Curr.Biol.2019, Exp.Neurobiol.2019, PNAS2018, Nat.Comm.2016

**Citations:** 1047 · h-index 13 · i10-index 17

as of July 2026

Suffixes: -w = workshop, -f = findings, -b = blogpost

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## ACHIEVEMENTS

**Seal of Excellence — MSCA Postdoctoral Fellowships 2025** · European Commission, Horizon Europe (HORIZON-MSCA-2025-PF-01-01) · 2026 *Proposal 101286042 — TRACE: Towards Reliable AI through Cognitive Engrams (with Max-Planck-Gesellschaft)*, total score 89.40%

**Excellent Paper Award** · Korean Artificial Intelligence Association (JKAIA) · 2023 *Brain-inspired Lp-Convolution benefits large kernels and aligns better with visual cortex*

**Patent Application & Technology Transfer** · Application Number: 10-2023-0027219 · 2023 *Device and Method for Classification of Human and Animal Behavior*, Patent Contribution=35%; Finalized a technology transfer with ACTNOVA Inc.

**Best Presentation Award** · Korean Society for Brain and Neural Sciences (KSBNS) · 2022 *ABCD-analysis: Mapping animal behavior and differential analysis of kinematic features without selection bias*

**Young Investigator Award** · Korean Society for Molecular and Cellular Biology (KSMCB) · 2022 *Retina-attached slice recording reveals light-triggered tonic GABA signaling in suprachiasmatic nucleus*

**Best Poster Award** · Center for Cognition and Sociality (CCS) Workshop · 2021 *Mapping Behavior with DECLARE: Deep Embedded Clustering of Action REpresentation*

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## TALKS

**AI Engram: In Search of Memory Traces in Artificial Neural Networks** · Scalable Trustworthy AI (STAI) group - KAIST · 2026 (Invited Seminar, In person, invited by Seong Joon Oh)

**AI Engram: In Search of Memory Traces in Artificial Neural Networks** · Max Planck Institute for Software Systems · 2026 (Invited Seminar, In person, invited by Krishna Gummadi)

**AI Engram: In Search of Memory Traces in Artificial Neural Networks** · Spark(l)ing Science - Max Planck Institute for Security and Privacy · 2026 (Seminar, In person)

**How neuroscience can benefit AI** · KAIST BK21 IBS Symposium - Integrated Neuroscience and Physiology · 2025 (Invited Seminar, In person, invited by Greg Suh)

**Transformer as a Hippocampal Memory Consolidation Model based on NMDAR-inspired Nonlinearity** · KSBNS Symposium · 2025 (Invited Seminar, In person)

**Sovereign AI: K-AI as an opportunity** · Netzwerk Junge Generation Deutschland-Korea · 2025 (Public Lecture, In person, invited by Muhong Lee)

**Brain-inspired AI** • Institute for Basic Science (IBS) • 2025 (Invited Seminar, In person, invited by C. Justin Lee)

**Brain-inspired AI** • Korea Brain Research Institute (KBRI) • 2025 (Invited Seminar, In person, invited by Jeongyeon Kim)

**Brain-inspired AI** • Gwangju Institute of Science and Technology (GIST) • 2025 (Invited Seminar, In person, invited by Sundong Kim)

**Neuroscience for Artificial Intelligence** • MLAI group, Yonsei University • 2025 (Invited Lecture, Offline, invited by Kyungwoo Song)

**Brain-inspired AI** • Kyungpook National University (KNU) • 2024 (Invited Seminar, In person, invited by Saekwang Nam)

**SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior** • Korea Institute of Ocean Science & Technology (KIOST) • 2024 (Invited Seminar, In person, invited by Soomee Kim)

**SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior** • Korea Brain Research Institute (KBRI) • 2024 (Invited Seminar, In person, invited by Jeongyeon Kim)

**Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity** • IBS-CCS Symposium • 2023 (Oral Presentation, In person)

**Memory editing in the brain - Eternal Sunshine of the Spotless Mind** • IBS Science in the Cinema Roof • 2023 (Public Lecture, In person)

**From human mind to artificial intelligence - Is ChatGPT a bridge** • Reading Festival at Jeonil High School • 2023 (Public Lecture, In person)

**Resolving clustering bias during animal behavior mapping and kinematic profiling** • IBS Data Science Group seminar • 2022 (Invited Seminar, Online, invited by Meeyoung Cha)

**Resemblances Between Transformer's Nonlinearity and NMDA Receptor Dynamics** • IBS Winter School on AI-Boosted Basic Science • 2022 (Invited Lecture, In person)

**Age-dependent motor coordination analysis by 3D pose-estimation using AVATAR** • IBS 2021 2nd Workshop • 2021 (Oral Presentation, In person)

**Backpropagation and the Brain** • KAIST-NeuroAI Journal Club • 2020 (Oral Presentation, Hybrid)

**Deep learning with NMDA receptor inspired activation function** • IBS 2019 Year-End Workshop • 2020 (Oral Presentation, In person)

**Development of a Low-cost, Comprehensive Recording System for Circadian Rhythm Behavior** • Korean Society for Molecular and Cellular Biology (KSMCB) • 2019 (Oral Presentation, In person)

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## TEACHING

**Guest Lecturer** • Georgia Tech CSE 8803 BMI "Brain-inspired Machine Intelligence", Fall 2024

**Guest Lecturer** • Ewha Womans University G 18067 "Introduction to System Health", Spring 2026

**Mentoring** • SoHyung Kim (Korea University, Undergraduate Intern), IBS 2020, 6 months • Outputs: CVPR2023-w, IJCV2024 (co-author)

**Mentoring** • Hyewon Kim (Eulji University School of Medicine, Undergraduate Intern), IBS 2024, 9 months • Outputs: ICLR2024-w (co-author), Exp.Neurobiol.2024 (co-first author)

**Mentoring** • Minsung Kim (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2024, 3 months • Outputs: ICLR2025-b, ACL2026 (first author)

**Mentoring** • Jiseon Kim (KAIST, Ph.D. Intern, with Luiz Felipe Vecchiatti), MPI-SP 2025, 6

months • Outputs: ICLR2025-w, ACL2026-f (first author)

**Mentoring** • Nakyeong Yang (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2025, 3 months • Outputs: ICLR2026 (first author)

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#### COMMUNITY SERVICE

**Symposium Organizer** • KSBNS 2026 Annual Meeting • "Minds Meet Machines: Bidirectional Innovations in AI and Neuroscience"

**Symposium Co-Organizer** • KSBNS & CJK 2025 Joint Symposium • "Neuroscience-inspired AI: Computational Insights into Biological and Artificial Intelligence"

**Program Committee** • AAAI 2026, AI Alignment Track

**Program Committee** • Machine+Behavior 2026, Harnack House, Berlin

**Emergency Reviewer** • ICML 2026

**Reviewer** • NeurIPS 2026

**Reviewer** • ICML 2026 Mechanistic Interpretability Workshop

**Reviewer** • ICLR 2026

**Reviewer** • ICLR 2026 Workshop on Representational Alignment (Re-Align)

**Reviewer** • ICLR 2025 Workshop on Representational Alignment (Re-Align)

**Founding Member** • KAIST-NeuroAI Journal Club, 2020

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#### PUBLICATIONS

✓ selected publication   \* first author   # corresponding author

- ✓ **J. Kwon\***, J. Kim, D. Kim, M. Cha. Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing. *ICML Mechanistic Interpretability Workshop*, 2026.
- ✓ **J. Kwon\***, D. Kim\*, J. Kim\*, Y. Kim, W. Kook, M. Cha. AI Engram: In Search of Memory Traces in Artificial Intelligence. *International Conference on Machine Learning*, 2026 (Oral).  
M. S. Locatelli, F. Tonucci, **J. Kwon**, L. F. Vecchietti, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha. Textual Supervision Enhances Geospatial Representations in Vision-Language Models. *International Conference on Machine Learning*, 2026 (Acceptance Rate=26.6%).
- ✓ J. Kim, **J. Kwon#**, L. F. Vecchietti#, W. Dong, J. Kim, M. Cha#. Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions. *Findings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=18%).  
M. Kim, D. Kim, **J. Kwon**, N. Yang, K. Jung, M. Cha. How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=19%).  
S. Kim, **J. Kwon**. From Human Cognition to AI Reasoning: Models, Methods, and Applications. *ICLR HCAIR Workshop*, 2026.
- ✓ **J. Kwon\***, S. Kim\*, J. Woo, K. Tanaka-Yamamoto, O. James, E. De Schutter, S. Hong, C. J. Lee. Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination. *Experimental & Molecular Medicine*, 2026 (5-year IF=14.2, 2024).
- ✓ **J. Kwon\***, L. F. Vecchietti, S. Park, M. Cha. Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 (AI Alignment Track, Acceptance Rate=22.65%).  
D. Kim, M. Kim, **J. Kwon**, N. Yang, M. Cha. Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).  
N. Yang, D. Kim, **J. Kwon**, M. Kim, K. Jung, M. Cha. Erase or Hide? Suppressing Spurious Un-

learning Neurons for Robust Unlearning. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).

M. Kim, **J. Kwon**, D. Kim, M. Cha. In Search of the Engram in LLMs: A Neuroscience Perspective on the Memory Functions in AI Models. *ICLR Blogposts Track*, 2025.

B. Hyeon, J. Shin, J. Lee, W. Kim, **J. Kwon**, H. Lee, D. Kim, C. Y. Kim, S. Choi, J. Jeong, K. Kim, C. J. Lee, D. Kim, W. D. Heo. Integrating Artificial Intelligence and Optogenetics for Parkinson's Disease Diagnosis and Therapeutics in Male Mice. *Nature Communications*, 2025.

J. Kim\*, **J. Kwon\***, L. F. Vecchietti\*, A. Oh, M. Cha. Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment. *ICLR Bi-Align Workshop*, 2025.

- ✓ **J. Kwon\***, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *International Conference on Learning Representations*, 2025 (Acceptance Rate=32.08%).

**J. Kwon\***, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in Identity-Aware Deep LSTM Network Encodes Obese-Like Motion Representations. *ICLR TS4H Workshop*, 2024.

**J. Kwon\***, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *ICLR Re-Align Workshop*, 2024.

- ✓ **J. Kwon\***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *International Journal of Computer Vision*, 2024 (5-year IF=15.5, 2024).

- ✓ **J. Kwon\***, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in a Deep Neural Network Encodes Obese-like Motion Representations. *Experimental Neurobiology*, 2024 (Featured on the cover).

H. Kang, A. Han, A. Zhang, H. Jeong, W. Koh, J. M. Lee, H. Lee, H. Y. Jo, M. A. Maria-Solano, M. Bhalla, **J. Kwon**, et al.. GolpHCat (TMEM87A), a Unique Voltage-Dependent Cation Channel in Golgi Apparatus, Contributes to Golgi-pH Maintenance and Hippocampus-Dependent Memory. *Nature Communications*, 2024.

- ✓ D. Kim\*, **J. Kwon\***, M. Cha, C. J. Lee. Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity. *Advances in Neural Information Processing Systems*, 2023 (Acceptance Rate=26.1%).

M. Sa, E. Yoo, W. Koh, M. G. Park, H. Jang, Y. R. Yang, M. Bhalla, J. Lee, J. Lim, W. Won, **J. Kwon**, et al.. Hypothalamic GABRA5-Positive Neurons Control Obesity via Astrocytic GABA. *Nature Metabolism*, 2023.

**J. Kwon\***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *CVPR CV4Animals Workshop*, 2023.

M. Nam, H. Y. Ko, D. Kim, S. Lee, Y. M. Park, S. J. Hyeon, W. Won, J. Chung, S. Y. Kim, H. H. Jo, K. T. Oh, Y. Han, G. Lee, Y. H. Ju, H. Lee, H. Kim, J. Heo, M. Bhalla, K. J. Kim, **J. Kwon**, et al.. Visualizing Reactive Astrocyte-Neuron Interaction in Alzheimer's Disease Using <sup>11</sup>C-Acetate and <sup>18</sup>F-FDG. *Brain*, 2023.

D. Kim, **J. Kwon**, M. Cha, C. J. Lee. Transformer Needs NMDA Receptor Nonlinearity for Long-Term Memory. *NeurIPS MemARI Workshop*, 2022.

D. Kim, J. Kim, W. Jung, J. Park, M. Kim, A. Shin, Y. Jeong, S. Park, G. Shin, Y. W. Lee, **J. Kwon**, D. Kim. AVATAR: AI Vision Analysis for Three-Dimensional Action in Real-Time. *CVPR CV4Animals Workshop*, 2022.

S. Kim, **J. Kwon**, M. G. Park, C. J. Lee. Dopamine-Induced Astrocytic Ca<sup>2+</sup> Signaling in mPFC is Mediated by MAO-B in Young Mice, but by Dopamine Receptors in Adult Mice. *Molecular Brain*, 2022.

Y. H. Ju, M. Bhalla, S. J. Hyeon, J. E. Oh, S. Yoo, U. Chae, **J. Kwon**, W. Koh, J. Lim, Y. M. Park, et al.. Astrocytic Urea Cycle Detoxifies A $\beta$ -Derived Ammonia While Impairing Memory in

Alzheimer's Disease. *Cell Metabolism*, 2022.

J. M. Lee, M. Sa, H. An, J. M. J. Kim, **J. Kwon**, B. Yoon, C. J. Lee. Generation of Astrocyte-Specific MAOB Conditional Knockout Mouse with Minimal Tonic GABA Inhibition. *Experimental Neurobiology*, 2022.

✓ **J. Kwon\***, M. W. Jang, C. J. Lee. Retina-Attached Slice Recording Reveals Light-Triggered Tonic GABA Signaling in Suprachiasmatic Nucleus. *Molecular Brain*, 2021.

M. W. Jang, T. Y. Kim, K. Sharma, **J. Kwon**, E. Yi, C. J. Lee. A Deafness Associated Protein TMEM43 Interacts with KCNK3 (TASK-1) Two-Pore Domain K<sup>+</sup> (K2P) Channel in the Cochlea. *Experimental Neurobiology*, 2021.

J. Won, H. H. Kazan, **J. Kwon**, M. Park, M. A. Ergun, S. Ozcan, B. Y. Choi, W. D. Heo, C. J. Lee. Ultimate COVID-19 Detection Protocol Based on Saliva Sampling and qRT-PCR with Risk Probability Assessment. *Experimental Neurobiology*, 2021.

K. Han, M. Lee, H. Lim, M. W. Jang, **J. Kwon**, C. J. Lee, S. Kim, M. Suh. Excitation-Inhibition Imbalance Leads to Alteration of Neuronal Coherence and Neurovascular Coupling Under Acute Stress. *Journal of Neuroscience*, 2020.

S. Oh, J. M. Lee, H. Kim, J. Lee, S. Han, J. Y. Bae, G. Hong, W. Koh, **J. Kwon**, E. Hwang, et al.. Ultrasonic Neuromodulation via Astrocytic TRPA1. *Current Biology*, 2019.

Y. Han, **J. Kwon**, J. Won, H. An, M. W. Jang, J. Woo, J. S. Lee, M. G. Park, B. Yoon, S. E. Lee, et al.. Tweety-Homolog (Ttyh) Family Encodes the Pore-Forming Subunits of the Swelling-Dependent Volume-Regulated Anion Channel (VRAC<sub>swell</sub>) in the Brain. *Experimental Neurobiology*, 2019.

J. Woo, J. O. Min, D. Kang, Y. S. Kim, G. H. Jung, H. J. Park, S. Kim, H. An, **J. Kwon**, J. Kim, et al.. Control of Motor Coordination by Astrocytic Tonic GABA Release Through Modulation of Excitation/Inhibition Balance in Cerebellum. *Proceedings of the National Academy of Sciences*, 2018.

**J. Kwon\***, M. G. Park, S. E. Lee, C. J. Lee. Development of a Low-Cost, Comprehensive Recording System for Circadian Rhythm Behavior. *Experimental Neurobiology*, 2018.

**J. Kwon\***, H. An, M. Sa, J. Won, J. I. Shin, C. J. Lee. Orai1 and Orai3 in Combination with Stim1 Mediate the Majority of Store-Operated Calcium Entry in Astrocytes. *Experimental Neurobiology*, 2017.

G. E. Ha, J. Lee, H. Kwak, K. Song, **J. Kwon**, S. Jung, J. Hong, G. Chang, E. M. Hwang, H. Shin, et al.. The Ca<sup>2+</sup>-Activated Chloride Channel Anoctamin-2 Mediates Spike-Frequency Adaptation and Regulates Sensory Transmission in Thalamocortical Neurons. *Nature Communications*, 2016.